

TrES-5b

R-0855

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240930 · created 2026-07-28T21:14:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460583.6875 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.036
Transit depth (Rp/Rs)^2	2.3 +/- 1.08 [%]
Orbital Inclination (inc)	83.52 +/- 2.29
Airmass coefficient 1 (a1)	1.096 +/- 0.027
Airmass coefficient 2 (a2)	-0.079 +/- 0.063
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	None
Optimal Aperture	2.67
Optimal Annulus	11.84
Transit Duration (day)	0.066 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.036 vs 0.142 (deviation 0.28σ)
Duration fit vs published	0.066 d vs 0.0758 d (deviation 0.55σ)
Mid-time O–C	4.4 min
Depth SNR	4.2
Residual scatter	2.46 %

Light curve

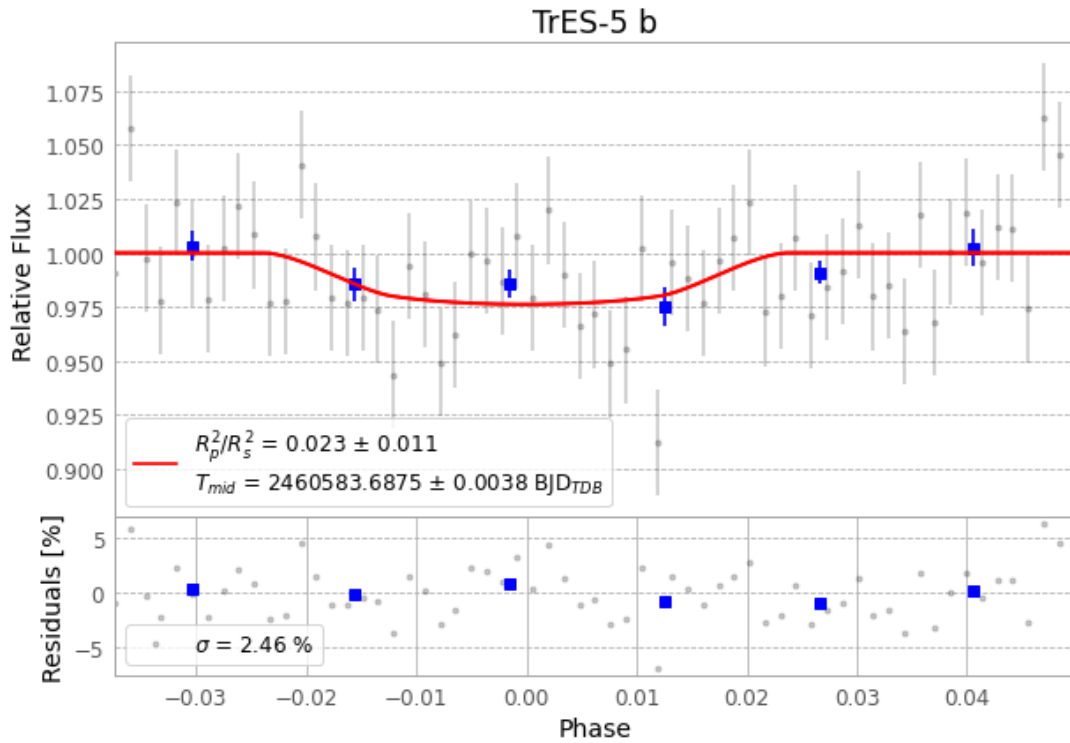


Figure 1 — FinalLightCurve_TrES-5 b_2024-09-29.png (SHA-256 943cc53474776ffc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0807d9e6ca6187a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

63 FITS frames (41 MB), TRES-5240930030617.FITS → TRES-5240930061215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240930.log (4453c953f30b...), FinalLightCurve_TrES-5 b_2024-09-29.png (943cc5347477...), FinalLightCurve_TrES-5 b_2024-09-29.csv (f0722d34bc23...)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:14Z.